Ames Housing Price Prediction — Final Report

Regression modeling of residential sale prices (Ames, Iowa dataset) — Linear Regression, Ridge, Lasso, KNN & SVR

1. Dataset & Preprocessing

- Raw data: 2,930 houses × 82 columns (Ames, Iowa housing dataset). Identifier columns (Order, PID) dropped; no duplicate rows found. Final feature set: 79 predictors + target (SalePrice).
- Features split into **34 numerical**, **25 nominal categorical** (one-hot encoded), and **20 ordinal categorical** (quality/condition scales, e.g. ExterQual, KitchenQual — encoded with explicit per-column rank orders, not a generic scale).
- Missing values: for 14 columns (BsmtQual, GarageType, PoolQC, FireplaceQu, etc.) NaN means "feature absent" per the data dictionary and was filled with the category "NA" before encoding. Remaining genuine missing values were imputed (median for numeric, constant "Missing" for categorical) inside the pipeline.
- Outlier removal: 3 houses with GrLivArea > 4,000 sq.ft. but SalePrice < $300k were dropped — a known Ames data quirk (likely partial/distressed sales) that would otherwise distort linear/SVR fits.
- Feature engineering: TotalSF (basement + 1st + 2nd floor area), HouseAge, RemodAge, and TotalBaths (weighted full/half baths) were engineered and added to the numeric feature set.
- Target & skewed numeric features log1p-transformed (|skew| > 0.75 rule) to correct right-skew; predictions are inverse-transformed (expm1) before reporting dollar metrics. All numeric/ordinal features scaled with StandardScaler (needed for distance-based KNN/SVR and scale-sensitive Ridge/Lasso penalties); one-hot columns left unscaled.
- Data split 60% train / 20% validation / 20% test (1,755 / 586 / 586 rows). Model selection used cross-validation + the validation set only; the test set was touched exactly once, for final reporting.

2. Key EDA Findings

- SalePrice is strongly right-skewed (skew ≈ 1.74; mean $180,796 vs. median $160,000, range $12,789–$755,000); log1p transform reduces skew to ≈ 0.05.
- OverallQual and GrLivArea show the strongest correlations with SalePrice and are consistently the top two predictors across every stage of the analysis (correlation heatmap, groupby analysis, and final coefficients).
- Neighborhood has a statistically significant relationship with price: one-way ANOVA F = 144.4 ($p \approx 0$); eta-squared = 0.573, i.e. neighborhood alone explains ~57% of price variance — a very large effect for a single categorical feature. NoRidge/StoneBr/NridgHt are the priciest neighborhoods; MeadowV/IDOTRR/BrDale the cheapest.
- Quality and size compound: the OverallQual × GrLivArea heatmap shows the highest average prices occur where both are simultaneously high, not from either alone.

3. Important Data Insights

- Houses with a basement sell for meaningfully more on average than those without.
- More bathrooms, more garage capacity, and higher kitchen quality are all associated with higher average SalePrice; central air conditioning also raises average price noticeably.
- House age is negatively correlated with price (older houses sell for less, on average), and sale month has only a mild effect on average price compared to structural/quality features.
- SaleCondition (e.g. Abnormal/foreclosure sales vs. Normal vs. Partial/new-construction sales) turned out to be an important price driver in the final model — a pattern not obvious from the univariate EDA alone.

4. Models Used

- Five baseline regressors, each inside an identical preprocessing pipeline (ColumnTransformer + model): **Linear Regression**, **Ridge** (alpha=10), **Lasso** (alpha=0.001), **KNN** (k=5), **SVR** (C=1, RBF kernel).
- 5-fold cross-validation (RMSE on log scale) was used to screen all five before committing time to tuning: Ridge and Lasso were the strongest baselines (CV RMSE ≈ 0.121), Linear Regression and SVR mid-table (≈ 0.146–0.148), KNN weakest (≈ 0.173).

5. Hyperparameter Tuning

- GridSearchCV (5-fold, same folds as model screening) was used to tune the two strongest/most tunable candidates — **Ridge** and **SVR**. Linear Regression has no hyperparameters; Lasso/KNN were already clearly behind and tuning them would not change the final choice.
- **Ridge grid:** alpha ∈ {0.01, 0.1, 1, 10, 20, 50} → best alpha = **50** (the largest value tried), i.e. the most regularized option generalized best — sensible given the large number of one-hot-encoded columns.
- **SVR grid:** C ∈ {0.1, 1, 10, 100}, gamma ∈ {'scale', 0.01, 0.1}, kernel = rbf → best C = **1**, gamma = **0.01**, i.e. tuning pushed toward a smoother, less complex function to correct the untuned model's overfitting.

- The winner between the two tuned candidates was chosen using validation-set RMSE only — the test set stayed untouched until the single final evaluation.

6. Evaluation Results

All metrics computed on the original dollar scale (log predictions inverse-transformed with expm1).

Model	CV RMSE (log)	Train R^2	Val R^2	Test R^2	Test RMSE	Test MAE
Linear Regression	0.1455	0.939	0.913	0.955	$19,651	$13,169
Ridge (default α=10)	0.1214	0.933	0.922	0.954	$19,361	$12,723
Lasso (α=0.001)	0.1211	0.921	0.921	0.948	$20,429	$13,302
KNN (k=5)	0.1728	0.882	0.832	0.843	$35,592	$20,765
SVR (default)	0.1481	0.956	0.866	0.904	$30,403	$15,708
Ridge (tuned, α=50)	0.1207	0.939	0.940	0.951	$19,822	$12,950
SVR (tuned, C=1, γ=0.01)	0.1422	-	0.839*	0.904*	-	-

*SVR (tuned) R^2 shown on log scale for val/test; RMSE/MAE omitted for space — see notebook Section 8 for the full table.

- **Overfitting:** SVR is the clearest case — default SVR train R^2=0.956 vs. validation R^2=0.866 (a large gap); tuning narrows but does not eliminate it. Ridge/Linear Regression/Lasso show only a small (~1–2 point) train-validation gap.
- **Underfitting:** KNN is weakest on every split (test R^2=0.843) — it doesn't fit the training data well to begin with, consistent with plain Euclidean distance struggling in a ~250-dimensional, mostly one-hot-encoded space.
- Tuning helped SVR meaningfully (validation R^2 up from 0.866 to ~0.88 on log scale) but only marginally improved Ridge, since the untuned default (α=10) was already close to optimal for this data.

7. Final Model Selection

Ridge Regression (tuned, α=50) is selected as the final model. Its train/validation/test R^2 (0.939 / 0.940 / 0.951) stay close together, showing it generalizes consistently — unlike SVR (large train-validation gap) or KNN (weak on every split). On the held-out test set it achieves an RMSE of **$19,822** and MAE of about **$12,950** ($R^2$ = 0.951 in dollar scale), meaning predictions are, on average, within roughly 7% of the actual sale price relative to the dataset's median ($160,000).

8. Main Conclusions

- Overall material/finish quality (OverallQual) and above-ground living area (GrLivArea) are the dominant drivers of Ames house prices, confirmed independently by EDA correlations, groupby analysis, and the final Ridge coefficients.
- Location matters almost as much as size/quality: Neighborhood alone explains ~57% of price variance (ANOVA eta-squared = 0.573), and specific neighborhoods (Crawford positive, IDOTRR negative) remain significant even after controlling for other features.
- The engineered TotalSF feature (combining basement + 1st + 2nd floor area) outperformed its individual components in the final model, validating the feature-engineering step.
- How a sale occurred (SaleCondition: normal vs. foreclosure/abnormal vs. new-construction/partial) is a meaningful, previously under-examined price factor that only emerged clearly from the modeling stage.
- A regularized linear model (Ridge) generalized more reliably than a more flexible non-linear model (SVR) on this feature space, showing that with ~250 mostly one-hot-encoded features, controlling variance via regularization mattered more than model flexibility.